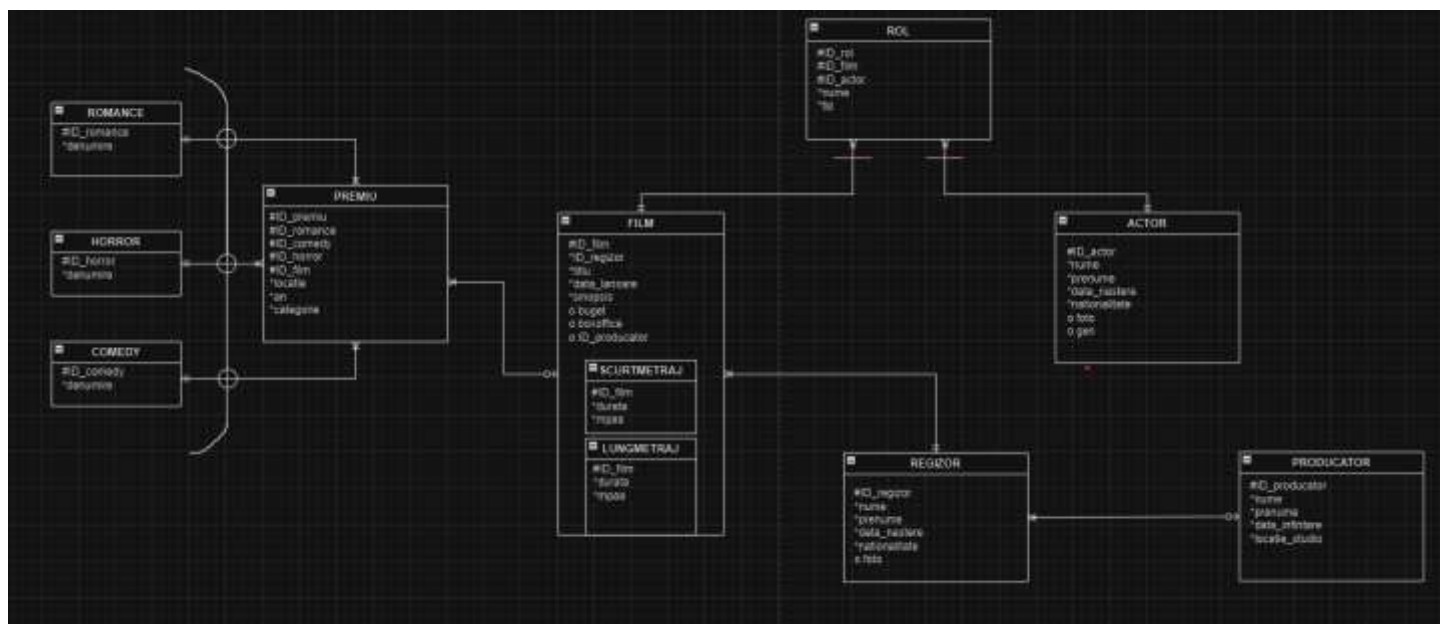


Proiect: Elemente Avansate de Baze de Date

Gestiunea Filmelor in scopul atribuirii unor premii

Filep Rebeca, IE An 3, Grupa 3, 2024-2025



TABELE DE MAPARE

SCURTMETRAJ			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK, FK	*	ID_film	
	*	durata	
	*	mpaa	

LUNGMETRAJ			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK, FK	*	ID_film	
	*	durata	
	*	mpaa	

ROL			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_rol	
FK	*	ID_film	Face legatura la filmul in care se afla rolul, printr-o relatie barata

FK	*	ID_actor	Face legatura la actorul care va avea rolul, printr-o relatie barata
	*	nume	Numele rolului/ personajului, ex.: Harap Alb
	*	tip	Acest tip se refera la: rol principal, rol secundar sau dublura (actor de rezerva)

ACTOR			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_actor	
	*	nume	
	*	prenume	
	*	data_nastere	
	*	nationalitate	→Tara unde are origine
	o	foto	URL catre poza
	o	gen	→Masculin sau feminin

REGIZOR			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_regizor	
	*	nume	
	*	prenume	
	*	data_nastere	
	*	nationalitate	
	o	foto	

PRODUCATOR			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_producator	
	*	nume	
	*	prenume	
	*	data_infintare	
	*	locatie_studio	Tara in care se afla studioul, compania de producatori de filme

PREMII			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_premiu	
FK	*	ID_romance	
FK	*	ID_horror	
FK	*	ID_comedy	
FK	*	ID_film	
	*	locatie	Denumirea festivalului de film in cadrul careia a fost dat un respective premiu
	*	an	
	*	categorie	Face referire la cele 3 FK-uri care sunt categoriile de film pentru care se poate primi premiu

HORROR			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_horror	
	*	denumire	Scena/tehnica/motivul pentru care a primit premiul, care a diferentiat filmul castigator de restul filmelor

ROMANCE			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_romance	
	*	denumire	

COMEDY			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_comedy	
	*	denumire	

FILM			OBSERVATII
Tip cheie	Optionalitate	Nume coloana	
PK	*	ID_film	
	*	titlu	
	*	data_lansare	
	*	sinopsis	
	o	buget	
	o	boxoffice	profit
FK	o	ID_producator	Este optional deoarece nu este obligatoriu ca orice film sa aiba o casa de productie care sa sponsorizeze
FK	*	ID_regizor	

SCHEMA BAZEI DE DATE NORMALIZATĂ

REGIZOR					
NUMBER(5)	VARCHAR2(100)	VARCHAR2(100)	VARCHAR2(100)	VARCHAR2(255)	DATE
id_regizor	nume	prenume	nationalitate	foto	data_nastere
11	Eggers	Robert	American	https://encrypted-	7/7/1983
12	Wes	Anderson	Englez	https://www.bifa.	2/15/1985
13	John	Doe	American	https://mediaprox	1/1/1970
14	King	Steven	American	https://stephenki	9/21/1947
15	Mkay	Adam	American	https://media.gq.	4/17/1968
16	Fey	Tina	American	https://www.imdb	2/2/1985

PRODUCATOR				
NUMBER(5)	VARCHAR2(100)	VARCHAR2(100)	VARCHAR2(255)	DATE
id_producator	nume	prenume	locatie_studio	data_infintare
21	A24	DanielKatz	New York	8/15/2012
22	Smith	Jane	Hollywood	2/2/1980
23	Lee	Samuel	Santa Maria	3/15/1965
24	Ephron	Nora	Chicago	5/19/1941
25	Mayers	Nancy	Los Angeles	12/8/1949
26				

ROL				
NUMBER(5)	NUMBER(5)2	NUMBER(5)3	VARCHAR2(255)	VARCHAR2(255)
id_rol	id_actor	id_film	nume	tip
301	401	111	Thomas	principal
302	501	111	Ephraim	principal
303	401	222	Christian	secundar
304	403	222	Yanis	principal
305	404	333	Cleopatra	secundar
306	405	333	Marius	secundar
307	406	444	Scream Queen	principal
308	407	444	Detective	secundar
309	408	555	Funny Single Man	Principal
310	409	555	Sidekick	secundar
310	409	555	Sidekick	secundar
310	409	555	Sidekick	secundar

Implementarea bazei de date direct cu constrangeri:

```
CREATE TABLE REGIZOR (  
ID_regizor NUMBER(5) PRIMARY KEY,  
nume VARCHAR2(100) NOT NULL,  
prenume VARCHAR2(100) NOT NULL,  
data_nastere DATE NOT NULL,  
nationalitate VARCHAR2(100) NOT NULL,  
foto VARCHAR2(255)  
);
```

```
CREATE TABLE PRODUCATOR (  
ID_producator NUMBER(5) PRIMARY KEY,  
nume VARCHAR2(100) NOT NULL,  
prenume VARCHAR2(100) NOT NULL,  
data_infiintare DATE NOT NULL,  
locatie_studio VARCHAR2(255)  
);
```

```
CREATE TABLE ACTOR (  
ID_actor NUMBER(5) PRIMARY KEY,  
nume VARCHAR2(100) NOT NULL,  
prenume VARCHAR2(100) NOT NULL,  
data_nastere DATE NOT NULL,  
nationalitate VARCHAR2(100) NOT NULL,  
foto VARCHAR2(255),  
gen CHAR(2)  
);
```

```
CREATE TABLE romance (  
ID_romance NUMBER(5) CONSTRAINT rmc_cif PRIMARY KEY,  
denumire VARCHAR2(255)  
);
```

```
CREATE TABLE horror (  
ID_horror NUMBER(5) CONSTRAINT hrr_cif PRIMARY KEY,  
denumire VARCHAR2(255)  
);
```

```
CREATE TABLE comedy (  
ID_comedy NUMBER(5) CONSTRAINT cmd_cif primary key,  
denumire VARCHAR2(255)  
);
```

```
CREATE TABLE FILM (  
ID_film NUMBER(5) PRIMARY KEY,  
ID_regizor NUMBER(5) NOT NULL,
```

```

        titlu VARCHAR2(255) NOT NULL,
        data_lansare DATE NOT NULL,
        sinopsis CLOB,
        buget NUMBER(10,2),
        boxoffice NUMBER(10,2),
        ID_producator NUMBER(5),
        FOREIGN KEY (ID_regizor) REFERENCES REGIZOR(ID_regizor),
        FOREIGN KEY (ID_producator) REFERENCES PRODUCATOR(ID_producator)
    );

```

```

CREATE TABLE ROL (
    ID_rol NUMBER(5) PRIMARY KEY,
    ID_film NUMBER(5) NOT NULL,
    ID_actor NUMBER(5) NOT NULL,
    nume VARCHAR2(255),
    tip VARCHAR2(255) NOT NULL,
    FOREIGN KEY (ID_film) REFERENCES FILM(ID_film),
    FOREIGN KEY (ID_actor) REFERENCES ACTOR(ID_actor)
);

```

```

CREATE TABLE SCURTMETRAJ (
    ID_film NUMBER(5) PRIMARY KEY,
    durata NUMBER(4,0),
    mpaa VARCHAR2(10),
    FOREIGN KEY (ID_film) REFERENCES FILM(ID_film)
);

```

```

CREATE TABLE LUNGMETRAJ (
    ID_film NUMBER(5) PRIMARY KEY,
    durata NUMBER(4,0),
    mpaa VARCHAR2(10),
    FOREIGN KEY (ID_film) REFERENCES FILM(ID_film)
);

```

```

CREATE TABLE PREMIU (
    ID_premiu NUMBER(5) PRIMARY KEY,
    ID_romance NUMBER(5),
    ID_comedy NUMBER(5),
    ID_horror NUMBER(5),
    ID_film NUMBER(5) NOT NULL,
    locatie VARCHAR2(255),
    an CHAR(10) NOT NULL,
    categorie VARCHAR2(255) NOT NULL,
    FOREIGN KEY (ID_romance) REFERENCES ROMANCE(ID_romance),
    FOREIGN KEY (ID_comedy) REFERENCES COMEDY(ID_comedy),
    FOREIGN KEY (ID_horror) REFERENCES HORROR(ID_horror),
    FOREIGN KEY (ID_film) REFERENCES FILM(ID_film),
    CHECK (
        (ID_romance IS NOT NULL AND ID_comedy IS NULL AND ID_horror IS NULL) OR
        (ID_romance IS NULL AND ID_comedy IS NOT NULL AND ID_horror IS NULL) OR

```



```
(ID_romance IS NULL AND ID_comedy IS NULL AND ID_horror IS NOT NULL)
)
);
```

Popularea tabelelor si adaugarea constrangerilor:

REGIZOR

```
INSERT INTO REGIZOR (ID_regizor, nume, prenume, data_nastere, nationalitate, foto)
VALUES (11, 'Eggers', 'Robert', TO_DATE('07/07/1983', 'DD/MM/YYYY'), 'American', 'https://encrypted-
tbn3.gstatic.com/licensed-image?q=tbn:ANd9GcQLgsnTPTDZNTWb5lgiRcMf-
CVeMW1RZJNerCYJ_q0qgbEGzsgPB4sWby91ahqecvrFXfKP2NQNaGaavEk');
```

```
INSERT INTO REGIZOR (ID_regizor, nume, prenume, data_nastere, nationalitate, foto)
VALUES (12, 'Wes', 'Anderson', TO_DATE('02/15/1985', 'DD/MM/YYYY'), 'Englez',
'https://www.bifa.film/people/benedict-cumberbatch/');
```

```
INSERT INTO REGIZOR (ID_regizor, nume, prenume, data_nastere, nationalitate, foto)
VALUES (13, 'John', 'Doe', TO_DATE('01/01/1970', 'DD/MM/YYYY'), 'American',
'https://www.bifa.film/people/benedict-cumberbatch/');
```

```
INSERT INTO REGIZOR (ID_regizor, nume, prenume, data_nastere, nationalitate, foto)
VALUES (13, 'King', 'Steven', TO_DATE('21/09/1947', 'DD/MM/YYYY'), 'American',
'https://stephenking.com/images/press/stephenking-sm.jpg');
```

```
INSERT INTO REGIZOR (ID_regizor, nume, prenume, data_nastere, nationalitate, foto)
VALUES (15, 'MKay', 'Adam', TO_DATE('17/04/1968', 'DD/MM/YYYY'), 'American',
'https://stephenking.com/images/press/stephenking-sm.jpg');
```

```
INSERT INTO REGIZOR (ID_regizor, nume, prenume, data_nastere, nationalitate, foto)
VALUES (16, 'Fey', 'Tina', TO_DATE('02/02/1985', 'DD/MM/YYYY'), 'American',
'https://www.imdb.com/name/nm0275486/mediaviewer/rm119144705/?ref_=ext_shr_lnk');
```

PRODUCATOR

```
INSERT INTO PRODUCATOR (ID_producator, nume, prenume, data_infiintare, locatie_studio)
VALUES (21, 'A24', 'DanielKatz', TO_DATE('2012-15-08', 'YYYY-MM-DD'), 'New York');
```

```
INSERT INTO PRODUCATOR (ID_producator, nume, prenume, data_infiintare, locatie_studio)
VALUES (22, 'Smith', 'Jane', TO_DATE('1980-02-02', 'YYYY-MM-DD'), 'Hollywood');
```

```
INSERT INTO PRODUCATOR (ID_producator, nume, prenume, data_infiintare, locatie_studio)
VALUES (23, 'Lee', 'Samuel', TO_DATE('1965-03-15', 'YYYY-MM-DD'), 'Santa Maria');
```

```
INSERT INTO PRODUCATOR (ID_producator, nume, prenume, data_infiintare, locatie_studio)
VALUES (24, 'Ephron', 'Nora', TO_DATE('1941-05-19', 'YYYY-MM-DD'), 'Chicago');
```

INSERT INTO PRODUCATOR (ID_producator, nume, prenume, data_infiintare, locatie_studio)
VALUES (25, 'Mayers', 'Nancy', TO_DATE('1949-08-12', 'YYYY-MM-DD'), 'Los Angeles');

ACTOR

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (401, 'Dafoe', 'Willem', TO_DATE('1955-07-22', 'YYYY-MM-DD'), 'American',
'https://www.hollywoodreporter.com/wp-content/uploads/2022/04/Willem-Dafoe-Northman-Premiere-Arrivals-GettyImages-1392276246-H-2022.jpg?w=1296&h=730&crop=1&resize=450%2C253', 'M');

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (402, 'Pattison', 'Robert', TO_DATE('1986-05-13', 'YYYY-MM-DD'), 'Englez',
'https://upload.wikimedia.org/wikipedia/commons/thumb/d/df/MJK_08789_Robert_Pattinson_%28Damsel%2C_Berlinale_2018%29_%28cropped%29.jpg/447px-MJK_08789_Robert_Pattinson_%28Damsel%2C_Berlinale_2018%29_%28cropped%29.jpg', 'M');

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (403, 'Cumberbatch', 'Benedict', TO_DATE('2002-05-19', 'YYYY-MM-DD'), 'Englez',
'https://www.bifa.film/people/benedict-cumberbatch/', 'M');

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (404, 'Blunt', 'Emily', TO_DATE('1983-02-23', 'YYYY-MM-DD'), 'Englez',
'https://static01.nyt.com/images/2022/07/24/business/22Retiring-Slush-01/22Retiring-Slush-01-mediumSquareAt3X.jpg', 'F');

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (405, 'Evans', 'Chris', TO_DATE('1981-06-13', 'YYYY-MM-DD'), 'American',
'https://images.unsplash.com/photo-1557862921-37829c790f19?q=80&w=1000&auto=format&fit=crop&ixlib=rb-4.0.3&ixid=M3wxMjA3fDB8MHxzZWVfY2h8M3x8bWFufGVufDB8fDB8fHww', 'M');

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (406, 'Popescian', 'Vio', TO_DATE('1958-11-22', 'YYYY-MM-DD'), 'Roman',
'https://media.meer.com/attachments/fab1026dcd503986811212d6bf4f44b28551991c/store/fill/860/645/95695d4e4c4394fb92cc829a02c7b4ed57dae7e1d0e9a5c2cf9600084c5e/A-bright-woman.jpg', 'F');

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (407, 'Willis', 'Bruce', TO_DATE('1955-03-19', 'YYYY-MM-DD'), 'Japonez',
'https://www.shutterstock.com/image-photo/portrait-young-handsome-man-smiling-600nw-213159280.jpg', 'M');

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (408, 'Will', 'Ferral', TO_DATE('1967-07-16', 'YYYY-MM-DD'), 'Scandinav',
'https://images.pexels.com/photos/874158/pexels-photo-874158.jpeg?cs=srgb&dl=pexels-olly-874158.jpg&fm=jpg', 'M');

INSERT INTO ACTOR (ID_actor, nume, prenume, data_nastere, nationalitate, foto, gen)
VALUES (409, 'Amy', 'Poehler', TO_DATE('1971-09-16', 'YYYY-MM-DD'), 'Italian',
'https://cms.forbesafrica.com/wp-content/uploads/2024/04/Amanda-Esther-Akokhia-started-her-first-photo-studio-at-the-age-of-24-image-supplied-scaled.jpg', 'F');

ROMANCE

```
INSERT INTO romance (ID_romance, denumire) VALUES (85, 'cu cele mai multe lacrimi');
INSERT INTO romance (ID_romance, denumire) VALUES (82, '0');
INSERT INTO romance (ID_romance, denumire) VALUES (83, '0');
INSERT INTO romance (ID_romance, denumire) VALUES (84, '0');
INSERT INTO romance (ID_romance, denumire) VALUES (85, '0');
```

COMEDY

```
INSERT INTO comedy (ID_comedy, denumire) VALUES (61, 'cea mai buna animatie');
INSERT INTO comedy (ID_comedy, denumire) VALUES (62, '0');
INSERT INTO comedy (ID_comedy, denumire) VALUES (63, '0');
INSERT INTO comedy (ID_comedy, denumire) VALUES (64, '0');
INSERT INTO comedy (ID_comedy, denumire) VALUES (65, '0');
```

HORROR

```
INSERT INTO horror (ID_horror, denumire) VALUES (71, 'Cel mai bun scenariu');
INSERT INTO horror (ID_horror, denumire) VALUES (72, 'Facut in cel mai scurt timp');
INSERT INTO horror (ID_horror, denumire) VALUES (73, '0');
INSERT INTO horror (ID_horror, denumire) VALUES (74, '0');
INSERT INTO horror (ID_horror, denumire) VALUES (75, '0');
```

FILM

```
INSERT INTO FILM (ID_film, ID_regizor, titlu, data_lansare, sinopsis, buget, boxoffice, ID_producator
) VALUES (111, 11, 'The LightHouse', TO_DATE('2019-10-18', 'YYYY-MM-DD'),
'https://en.wikipedia.org/wiki/The_Lighthouse_(2019_film)', '11', '18.3', 21);
```

```
INSERT INTO FILM (ID_film, ID_regizor, titlu, data_lansare, sinopsis, buget, boxoffice, ID_producator
) VALUES (222, 12, 'The wonderful story of Henry Sugar', TO_DATE('2023-09-20', 'YYYY-MM-DD'),
'https://en.wikipedia.org/wiki/The_Wonderful_Story_of_Henry_Sugar_(film)', '3', '5', 25);
```

Si altele

ROL

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (301, 401, 111, 'Thomas', 'Principal');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (302, 403, 111, 'Ephraim', 'Principal');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (303, 402, 222, 'Christian', 'Secundar');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (304, 403, 222, 'Yanis', 'Principal');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (305, 404, 333, 'Cleopatra', 'Secundar');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (306, 405, 333, 'Marius', 'Secundar');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (307, 406, 444, 'Scream Queen', 'Principal');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (308, 407, 444, 'Detective', 'Secundar');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (309, 408, 555, 'Funny single man', 'Principal');
```

```
INSERT INTO ROL (ID_rol, ID_actor, ID_film, nume, tip)
VALUES (310, 409, 555, 'Sidekick', 'Secundar');
```

SCURTMETRAJ

```
INSERT INTO SCURTMETRAJ (ID_film, durata, mpaa)
VALUES (222, 39, 'PG-13');
```

```
INSERT INTO SCURTMETRAJ (ID_film, durata, mpaa)
VALUES (333, 50, 'PG-13');
```

LUNGMETRAJ

```
INSERT INTO LUNGMETRAJ (ID_film, durata, mpaa)
VALUES (111, 110, 'R');
```

```
INSERT INTO LUNGMETRAJ (ID_film, durata, mpaa)
VALUES (444, 130, 'PG-13');
```

```
INSERT INTO LUNGMETRAJ (ID_film, durata, mpaa)
VALUES (555, 90, 'PG-13');
```

PREMIU

```
INSERT INTO PREMIU (
```

```

ID_premiu, ID_romance, ID_comedy, ID_horror, ID_film, locatie, an, categorie
) VALUES (
51, NULL, NULL, 71, 111, 'CANNES','2019', 'horror');

INSERT INTO PREMIU (
ID_premiu, ID_romance, ID_comedy, ID_horror, ID_film, locatie, an, categorie
) VALUES (
52, NULL, 61, NULL, 222, 'OHIO','2024', 'comedy');

INSERT INTO PREMIU (
ID_premiu, ID_romance, ID_comedy, ID_horror, ID_film, locatie, an, categorie
) VALUES (
53, 81, NULL, NULL, 333, 'Los Angeles','2024', 'romance');

INSERT INTO PREMIU (
ID_premiu, ID_romance, ID_comedy, ID_horror, ID_film, locatie, an, categorie
) VALUES (
54, NULL, NULL, 72, 444, 'New York','2005', 'horror');

INSERT INTO PREMIU (
ID_premiu, ID_romance, ID_comedy, ID_horror, ID_film, locatie, an, categorie
) VALUES (
55, NULL, 62, NULL, 555, 'TIFF','2018', 'comedy');

```

Packages

a.) Crearea/specificatia pachetului

Pachetul creat este **filme_regizori**. Am declarat o variabila publica “v_max_budget” care este initiata cu valoarea 0, iar la nevoie isi schimba valoarea cu cea mai mare valoare a bugetului inregistrata pentru un film prin functia “GetMaxBuget”. Mai avem 2 proceduri publice “AdaugaFilm” si “ModificaFilm”, 3 functii publice (‘GetBoxOfficeFilm’, ‘GetActoriFilme’, ‘GetRegizoriSiFilme’) si cursori creati folosind table of records(‘t_lista_actori’, ‘t_lista_filme’) si unul static, respectiv fara parametrii (‘c_boxoffice_filme’).

```

CREATE OR REPLACE PACKAGE filme_regizori AS
v_max_budget NUMBER := 0; //variabila publica

FUNCTION GetMaxBuget RETURN NUMBER;

```

```

//Proceduri publice
PROCEDURE AdaugaFilm(

```

```
p_id_film NUMBER,  
p_id_regizor NUMBER,  
p_titlu VARCHAR2,  
p_data_lansare DATE,  
p_sinopsis CLOB,  
p_buget NUMBER,  
p_boxoffice NUMBER,  
p_id_producator NUMBER  
);
```

```
PROCEDURE ModificaFilm(  
    p_id_film NUMBER,  
    p_titlu VARCHAR2  
);
```

```
TYPE t_actor IS RECORD (  
    id_actor NUMBER,  
    nume VARCHAR2(100),  
    prenume VARCHAR2(100)  
);
```

```
TYPE t_film IS RECORD (  
    id_film NUMBER,  
    titlu VARCHAR2(255),  
    id_regizor NUMBER,  
    nume_regizor VARCHAR2(100),  
    prenume_regizor VARCHAR2(100)  
);
```

//cursorul multiplu si cel explicit cu parametri

```
TYPE ref_cursor_filme IS REF CURSOR RETURN t_film;  
TYPE ref_cursor_actors IS REF CURSOR RETURN t_actor;
```

//Functii publice

```
FUNCTION GetBoxOfficeFilm RETURN VARCHAR2; //filmul cu cele mai multe incasari  
FUNCTION GetActoriFilme(p_id_film NUMBER) RETURN t_lista_actors; // Lista cu toti actorii care  
joaca in filmul oferit ca parametru  
FUNCTION GetRegizoriSiFilme RETURN t_lista_filme; // Lista cu toate filmele si regizorii asociati
```

```
CURSOR c_boxoffice_filme IS  
    SELECT titlu, boxoffice FROM FILM; //cursor fara parametrul (static)
```

```
END filme_regizori;
```

b.) Corpul Pachetului filme_regizori

In corpul pachetului avem implementate detaliile interne. Se regaseste implementarea functiilor si procedurilor publice, cat si declararea si implementarea functiei si procedurilor private. Spre exemplu functia privata “GetNumarActoriFilm” care verifica (in ‘GetActoriFilme’) daca un film a atribuit roluri in baza de date pentru filmul pe care l-a adaugat.

```
CREATE OR REPLACE PACKAGE BODY filme_regizori AS
```

```
FUNCTION GetMaxBuget RETURN NUMBER IS
```

```
    v_max NUMBER;
```

```
BEGIN
```

```
    SELECT NVL(MAX(buget), 0) INTO v_max FROM FILM;
```

```
    RETURN v_max;
```

```
END GetMaxBuget;
```

```
PROCEDURE AdaugaFilm(
```

```
    p_id_film NUMBER,
```

```
    p_id_regizor NUMBER,
```

```
    p_titlu VARCHAR2,
```

```
    p_data_lansare DATE,
```

```
    p_sinopsis CLOB,
```

```
    p_buget NUMBER,
```

```
    p_boxoffice NUMBER,
```

```
    p_id_producator NUMBER
```

```
) IS
```

```
BEGIN //Procedura privata care verifica bugetul sa fie mai mic decat maximul
```

```
    v_max_budget := GetMaxBuget();
```

```
    IF p_buget > v_max_budget THEN
```

```
        DBMS_OUTPUT.PUT_LINE('Atenție! Bugetul introdus este mai mare decât cel mai mare buget  
înregistrat.');
```

```
    END IF;
```

```
    INSERT INTO FILM (ID_film, ID_regizor, titlu, data_lansare, sinopsis, buget, boxoffice,  
ID_producator) //cursor implicit(insert)
```

```
    VALUES (p_id_film, p_id_regizor, p_titlu, p_data_lansare, p_sinopsis, p_buget, p_boxoffice,  
p_id_producator);
```

```
    v_max_budget := GetMaxBuget();
```

```
    DBMS_OUTPUT.PUT_LINE('Filmul a fost adăugat cu succes! Bugetul maxim este acum: ' ||  
v_max_budget);
```

```
END AdaugaFilm;
```

```
PROCEDURE ModificaFilm(
```

```

    p_id_film NUMBER,
    p_titlu VARCHAR2
) IS
BEGIN
    UPDATE FILM
    SET titlu = p_titlu
    WHERE ID_film = p_id_film;

    IF SQL%ROWCOUNT = 0 THEN
        RAISE_APPLICATION_ERROR(-20003, 'Filmul nu a fost găsit!');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Filmul a fost modificat cu succes!');
    END IF;
END ModificaFilm;

```

```

FUNCTION GetRegizoriSiFilme RETURN ref_cursor_filme IS
    rc ref_cursor_filme;
BEGIN
    OPEN rc FOR
        SELECT F.ID_film, F.titlu, R.ID_regizor, R.num, R.prenume
        FROM FILM F
        JOIN REGIZOR R ON F.ID_regizor = R.ID_regizor
        ORDER BY R.num, F.titlu;
    RETURN rc;
END GetRegizoriSiFilme;

```

// Functie privata folosita in GetActoriFilme

```

FUNCTION GetNumarActoriFilm(p_id_film NUMBER) RETURN NUMBER IS
    v_count NUMBER;
BEGIN
    SELECT COUNT(*) INTO v_count
    FROM ROL
    WHERE ID_film = p_id_film;

    RETURN v_count;
END GetNumarActoriFilm;

```

```

FUNCTION GetActoriFilme(p_id_film NUMBER) RETURN ref_cursor_actors IS
    rc ref_cursor_actors;
    v_numar_actors NUMBER;
BEGIN
    v_numar_actors := GetNumarActoriFilm(p_id_film);

    IF v_numar_actors = 0 THEN
        RAISE_APPLICATION_ERROR(-2005, 'Filmul nu are rolurile distribuite');
    END IF;

    OPEN rc FOR
        SELECT A.ID_actor, A.num, A.prenume

```



```

FROM ROL R
JOIN ACTOR A ON R.ID_actor = A.ID_actor
WHERE R.ID_film = p_id_film
ORDER BY A.num, A.prenume;

RETURN rc;
END GetActoriFilme;

FUNCTION GetBoxOfficeFilm RETURN VARCHAR2 IS
  v_titlu VARCHAR2(255);
  v_boxoffice NUMBER := 0;
  v_temp_titlu VARCHAR2(255);
  v_temp_boxoffice NUMBER;
BEGIN
  OPEN c_boxoffice_filme;
  LOOP
    FETCH c_boxoffice_filme INTO v_temp_titlu, v_temp_boxoffice;
    EXIT WHEN c_boxoffice_filme%NOTFOUND;

    IF v_temp_boxoffice > v_boxoffice THEN
      v_titlu := v_temp_titlu;
      v_boxoffice := v_temp_boxoffice;
    END IF;
  END LOOP;
  CLOSE c_boxoffice_filme;
  RETURN 'Cel mai profitabil film: ' || v_titlu || ' (Încasări: ' || v_boxoffice || ')';
END GetBoxOfficeFilm;

END filme_regizori;

```

C.) Triggers

- Trigger-ul ‘trg_protectie_regizor’ asigura faptul ca daca avem filme asociate unui regizor existente in baza de data atunci un regizor nu poate fi sters.

```

CREATE OR REPLACE TRIGGER trg_protectie_regizor
BEFORE DELETE ON REGIZOR
FOR EACH ROW
DECLARE
  v_count NUMBER;
BEGIN
  SELECT COUNT(*) INTO v_count FROM FILM WHERE ID_regizor = :OLD.ID_regizor;

  IF v_count > 0 THEN
    RAISE_APPLICATION_ERROR(-20010, 'Regizorul nu poate fi şters deoarece are filme asociate.');
```

END;

- Trigger-ul trg_verificare_producator este folosit ca un fel de conditie care asigura asocierea unui producator in cazul filmelor care au avut un castig extrem de mare (Box Office) pentru a permite incadrarea lor in mai multe posibile categorii de premii.

```
CREATE OR REPLACE TRIGGER trg_verifica_producator
BEFORE INSERT OR UPDATE OF boxoffice ON FILM
FOR EACH ROW
DECLARE
    v_max_boxoffice NUMBER;
BEGIN
    SELECT MAX(boxoffice) INTO v_max_boxoffice FROM FILM;

    IF :NEW.boxoffice > v_max_boxoffice AND :NEW.ID_producator IS NULL THEN
        RAISE_APPLICATION_ERROR(-20020, '✗ Nu poți insera/modifica acest film fără producător asociat!');
    END IF;
END;
```

D.) Secventa seg_film_id care asigura ca fiecare film introdus va avea un film_id unic care urmeaza tiparul de 3 cifre identice

```
CREATE SEQUENCE seq_film_id
START WITH 111
INCREMENT BY 111
MAXVALUE 999
CYCLE
NOCACHE;
```

Bloc PL/SQL:

```
DECLARE
    v_id_film NUMBER(3);
    v_count NUMBER;
BEGIN
    LOOP
        SELECT seq_film_id.NEXTVAL INTO v_id_film FROM DUAL;

        SELECT COUNT(*) INTO v_count FROM FILM WHERE ID_FILM = v_id_film;

        IF v_count = 0 THEN
            EXIT;
        END IF;
    END LOOP;
```

```

INSERT INTO FILM (ID_FILM, ID_REGIZOR, TITLU, DATA_LANSARE, SINOPSIS, BUGET,
BOXOFFICE, ID_PRODUCATOR)
VALUES (v_id_film, 15, 'Before Sunset', TO_DATE('2004-07-02', 'YYYY-MM-DD'),
'What if you had a second chance with the one that got away?', 2.7, 15, 25);

```

```

DBMS_OUTPUT.PUT_LINE('☑ Film adăugat cu ID-ul: ' || v_id_film);
END;

```

The screenshot shows a SQL IDE with a dark theme. The top toolbar contains icons for undo, redo, search, and a pin. The main editor area displays a PL/SQL script with line numbers 1 through 14. The script declares two variables, enters a loop to generate a new ID and check for its existence, and exits if it already exists. Below the editor, a tabbed interface shows the 'Results' tab selected, displaying a green checkmark, a confirmation message, and execution statistics.

```

1 DECLARE
2     v_id_film NUMBER(3);
3     v_count NUMBER;
4 BEGIN
5     LOOP
6         SELECT seq_film_id.NEXTVAL INTO v_id_film FROM DUAL;
7
8         SELECT COUNT(*) INTO v_count FROM FILM WHERE ID_FILM = v_id_film;
9
10        IF v_count = 0 THEN
11            EXIT;
12        END IF;
13    END LOOP;
14
Results Explain Describe Saved SQL History
☑ Film adăugat cu ID-ul: 777
1 row(s) inserted.
0.00 seconds

```

MongoDB

- Scenariul pentru acest proiect presupune o baza de date care se concentreaza pe 2 elemente de baza: Filme si Actori. Cele 3 colectii pe care le contine sunt Film, Actor si Rol. O posibila aplicatie ofera utilizatorului o vizualizare simpla si rapida a infomatiilor legate de un film, de la ce buget a avut inainte de a fi filmat, pana la tara din care provine actorul principal, totul in contextul unui singur film preferat.
- **Colectia principala “Film”** va contine detalii despre cate un film in parte si detalii despre Regizor si Actori imbricate. **Colectia “Rol”** leaga actorii de filme ca si o tabela de intersectie, specificand **ce rol** a avut **ce actor** in **ce film**, deoarece un actor poate juca in mai multe filme, dar poate avea si 2 roluri in acelasi film. **Colectia “Actor”** contine doar informatii despre actor, pentru a evita duplicarea informatiilor din Film.

- Utilizarea documentelor imbricate pentru Regizor si Actor in Film:

- > Pentru a simplifica accesul la date am decis stocarea regizorului direct in film, deoarece este unic pentru fiecare film, evitand creerea unei noi colectii 'Regizor'.

- > Informatiile imbricate legate de regizor si actori sunt id-urile si numele lor pentru a oferi un acces rapid la detaliile despre orice film. Colectia 'Film' contine de asemenea informatii despre id-ul filmului, titlu, data in care a fost lansat si care a fost profitul facut (Box-Office).

- Mentinerea colectiilor separate:

- > Teoretic am putea avea actorii in colectia 'Film' dar ar fi dificil de actualizat cand un actor are nou rol, devenind redundant sa repetam aceleasi date despre el din nou si din nou. Iar daca am pastra detaliile despre roluri in 'Film', ar complica interogari precum "In cate filme joaca actorul x?". Datorita faptului ca avem colectia Rol care reprezinta relatia Actor-Film, putem sa colectam toate rolurile intr-o singura interogare.

Crearea bazei de date si a colectiilor

```
0% mongosh mongodb://127.0.0.1:27017/?directCo
FilmeDB> use FilmDB
switched to db FilmDB
FilmeDB> db.createCollection("Film")
{ ok: 1 }
FilmeDB> db.createCollection("Rol")
{ ok: 1 }
FilmeDB> db.createCollection("Actor")
{ ok: 1 }
FilmeDB>
```

- Inserarea datelor in colectia Actor:

```
{
  "_id": 401,
  "nume": "Dafoe",
  "prenume": "Willem",
  "nationalitate": "American",
  "data_nastere": "1995-07-22",
  "gen": "M"
}
```

```
{
  "_id": 402,
  "nume": "Pattison",
  "prenume": "Robert",
  "nationalitate": "Englez",
  "data_nastere": "1986-05-13",
}
```

```
    "gen": "M"  
  }
```

```
{  
  "_id": 403,  
  "nume": "Cumberbatch",  
  "prenume": "Benedict",  
  "nationalitate": "Englez",  
  "data_nastere": "2002-05-19",  
  "gen": "M"  
}
```

```
{  
  "_id": 404,  
  "nume": "Blunt",  
  "prenume": "Emily",  
  "nationalitate": "Englez",  
  "data_nastere": "1983-02-23",  
  "gen": "F"  
}
```

```
{  
  "_id": 405,  
  "nume": "Evans",  
  "prenume": "Chris",  
  "nationalitate": "American",  
  "data_nastere": "1981-06-13",  
  "gen": "M"  
}
```

```
{  
  "_id": 406,  
  "nume": "Popescian",  
  "prenume": "Vio",  
  "nationalitate": "Roman",  
  "data_nastere": "1958-11-22",  
  "gen": "M"  
}
```

```
{  
  "_id": 407,  
  "nume": "Bruce",  
  "prenume": "Willis",  
  "nationalitate": "Japonez",  
  "data_nastere": "1955-03-19",  
  "gen": "M"  
}
```

```
{  
  "_id": 408,  
  "nume": "Will",
```

```

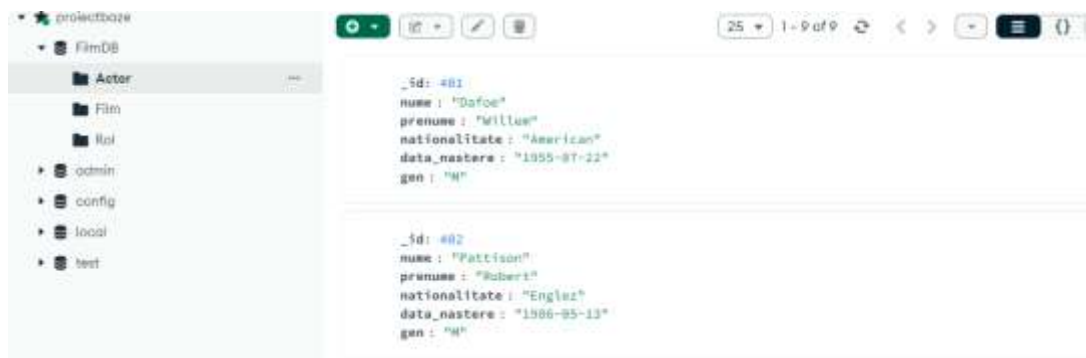
"prenume": "Ferral",
"nationalitate": "Scandinav",
"data_nastere": "1967-07-16",
"gen": "M"
}

```

```

{
  "_id": 409,
  "nume": "Amy",
  "prenume": "Poehler",
  "nationalitate": "Italianca",
  "data_nastere": "1971-09-16",
  "gen": "F"
}

```



● Inserarea datelor in colectia Film:

```

{
  "titlu": "The Lighthouse",
  "regizor": {
    "id_regizor": 11,
    "nume": "Robert Eggers"
  },
  "actori": [
    {
      "id_actor": 401,
      "nume": "Willem Dafoe"
    },
    {
      "id_actor": 402,
      "nume": "Robert Pattinson"
    }
  ],
  "data_lansare": "2019-10-18",
  "boxoffice": 18.3
}

```

```

{

```

```
"titlu": "The Wonderful story of Henry Sugar",
"regizor": {
  "id_regizor": 12,
  "nume": "Wes Anderson"
},
"actori": [
  {
    "id_actor": 401,
    "nume": "Willem Dafoe"
  },
  {
    "id_actor": 403,
    "nume": "Cumberbatch Benedict"
  }
],
"data_lansare": "2023-09-20",
"boxoffice": 5
}
```

```
{
  "titlu": "The Great Adventure",
  "regizor": {
    "id_regizor": 13,
    "nume": "Fey Tina"
  },
  "actori": [
    {
      "id_actor": 404,
      "nume": "Blunt Emily"
    },
    {
      "id_actor": 405,
      "nume": "Evans Chris"
    }
  ],
  "data_lansare": "2024-05-31",
  "boxoffice": 15
}
```

```
{
  "titlu": "Haunted Night",
  "regizor": {
    "id_regizor": 14,
    "nume": "Steven King"
  },
  "actori": [
    {
      "id_actor": 406,
      "nume": "Popescian Vio"
    },
    {

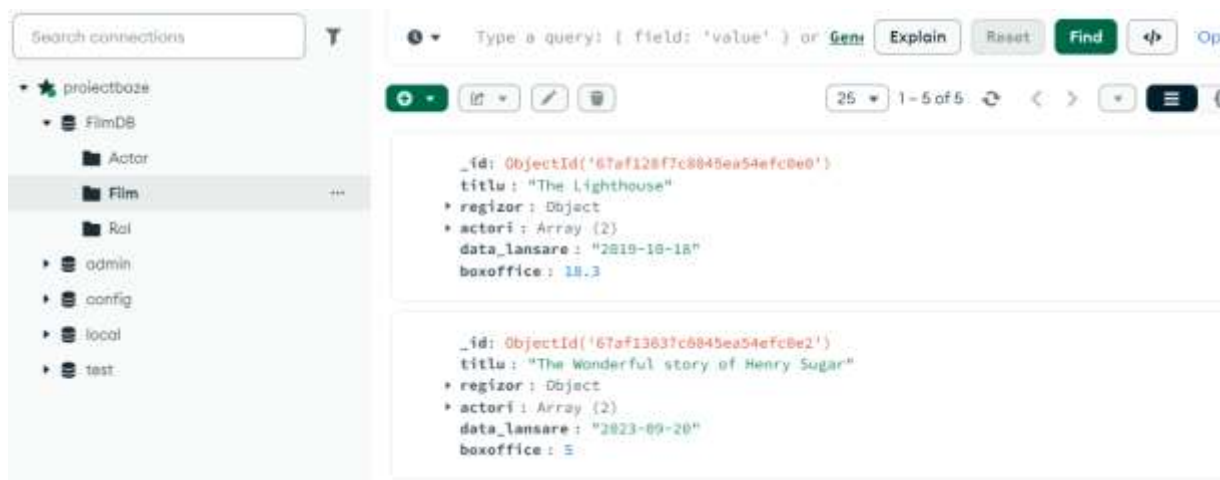
```

```

      "id_actor": 407,
      "nume": "Bruce Willis"
    },
  ],
  "data_lansare": "2005-10-31",
  "boxoffice": 5
}

{
  "titlu": "Laugh Out Loud",
  "regizor": {
    "id_regizor": 15,
    "nume": "MKay Adam"
  },
  "actori": [
    {
      "id_actor": 408,
      "nume": "Will Ferral"
    },
    {
      "id_actor": 409,
      "nume": "Amy Poehler"
    }
  ],
  "data_lansare": "2018-07-04",
  "boxoffice": 12
}

```



● Inserarea datelor in colectia Rol:

```

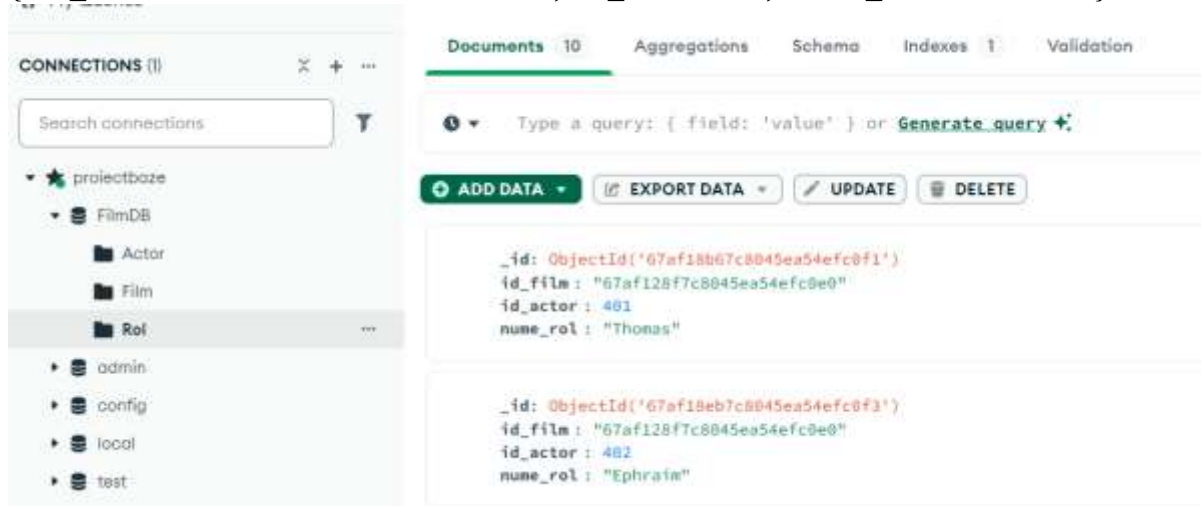
{ "id_film": "67af128f7c8045ea54efc0e0", "id_actor": 401, "nume_rol": "Thomas" }
{ "id_film": "67af128f7c8045ea54efc0e0", "id_actor": 402, "nume_rol": "Ephraim" }
{ "id_film": "67af13837c8045ea54efc0e2", "id_actor": 401, "nume_rol": "Christian" }

```



```
{ "id_film": "67af13837c8045ea54efc0e2", "id_actor": 403, "nume_rol": "Yanis" }
{ "id_film": "67af145b7c8045ea54efc0e4", "id_actor": 404, "nume_rol": "Cleopatra" }
{ "id_film": "67af145b7c8045ea54efc0e4", "id_actor": 405, "nume_rol": "Marius" }
{ "id_film": "67af14f07c8045ea54efc0e6", "id_actor": 406, "nume_rol": "Scream Queen" }
{ "id_film": "67af14f07c8045ea54efc0e6", "id_actor": 407, "nume_rol": "Detective" }

{ "id_film": "67af15c17c8045ea54efc0e8", "id_actor": 408, "nume_rol": "Funny Single Man" }
{ "id_film": "67af15c17c8045ea54efc0e8", "id_actor": 409, "nume_rol": "Sidekick" }
```



- Operatia de stergere

```
Microsoft Windows [Version 10.0.19045.5487]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Rebecca Files>mongo
Current MongoDB Log ID: 67af126d76f30fda00d318f
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongo
2.3.3
Using MongoDB: 8.0.3
Using Mongosh: 2.3.3
Mongosh 2.3.9 is available for download: https://www.mongodb.com/try/download/shell
For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2025-02-13T22:16:39.400+02:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use FilmDB
switched to db FilmDB
filmDB> db.Film.deleteOne({"title": "Laugh Out Loud"});
{ acknowledged: true, deletedCount: 1 }
filmDB> db.Film.find({"title": "Laugh Out Loud"});
filmDB>
```

- Operatia de actualizare

```
FilmDB> db.Film.updateOne(
... {"titlu": "The Great Adventure" }, {$set: {"boxoffice": 17} }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
FilmDB> db.Film.find({"titlu": "The Great Adventure"});
[
  {
    _id: ObjectId('67af145b7c8b45ea54efc0e4'),
    titlu: 'The Great Adventure',
    regizor: { id_regizor: 13, nume: 'Fey Tina' },
    actori: [
      { id_actor: 404, nume: 'Blunt Emily' },
      { id_actor: 405, nume: 'Evans Chris' }
    ],
    data_lansare: '2024-05-31',
    boxoffice: 17
  }
]
FilmDB>
```

- Indexul Compus

```
FilmDB> db.Film.createIndex({"titlu": 1, "boxoffice": -1 });
titlu_1_boxoffice_-1
FilmDB> db.Film.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { titlu: 1, boxoffice: -1 },
    name: 'titlu_1_boxoffice_-1'
  }
]
FilmDB>
```

- Interogari:

- 1) Filme lansate dupa 2010

Folosind 'gt'(greater than) va gasi toate filmele care au variabila data_lansare mai mare de 2010 si le va afisa:

```
FilmDB> db.Film.find({"data_lansare": { $gt: ISODate("2010-01-01") } });
[
  {
    _id: ObjectId('67af128f7c8045ea54efc0e0'),
    titlu: 'The Lighthouse',
    regizor: { id_regizor: 11, nume: 'Robert Eggers' },
    actori: [
      { id_actor: 401, nume: 'Willem Dafoe' },
      { id_actor: 402, nume: 'Robert Pattinson' }
    ],
    data_lansare: ISODate('2019-10-18T00:00:00.000Z'),
    boxoffice: 18.3
  },
  {
    _id: ObjectId('67af13037c8045ea54efc0e2'),
    titlu: 'The Wonderful story of Henry Sugar',
    regizor: { id_regizor: 12, nume: 'Wes Anderson' },
    actori: [
      { id_actor: 401, nume: 'Willem Dafoe' },
      { id_actor: 403, nume: 'Cumberbatch Benedict' }
    ],
    data_lansare: ISODate('2023-09-20T00:00:00.000Z'),
    boxoffice: 5
  },
  {
    _id: ObjectId('67af145b7c8045ea54efc0e4'),
    titlu: 'The Great Adventure',
    regizor: { id_regizor: 13, nume: 'Fey Tina' },
    actori: [
      { id_actor: 404, nume: 'Blunt Emily' },
      { id_actor: 405, nume: 'Evans Chris' }
    ],
    data_lansare: ISODate('2024-05-31T00:00:00.000Z'),
    boxoffice: 17
  }
]
```

Activate Windows

2) Cautarea unui actor dupa prenume

Va gasi toti actorii care au acelasi prenume si va afisa datele despre ei:

```
FilmDB> db.Actor.find({"prenume": "Robert"});
[
  {
    _id: 402,
    nume: 'Pattison',
    prenume: 'Robert',
    nationalitate: 'Englez',
    data_nastere: '1986-05-13',
    gen: 'M'
  }
]
FilmDB>
```

3) Ordonarea filmelor descrescator

Cauta in documentele din Colectia Film, filmele care au boxoffice mai mare de 10 si le afiseaza in ordine descrescatoare.

```
FilmDB> db.Film.find(
... { "boxoffice": { $gt: 10 } }
... ).sort(
... { "boxoffice": -1 } ).pretty();
[
  {
    _id: ObjectId('67af128f7c8045ea54efc0e0'),
    titlu: 'The Lighthouse',
    regizor: { id_regizor: 11, nume: 'Robert Eggers' },
    actori: [
      { id_actor: 401, nume: 'Willem Dafoe' },
      { id_actor: 402, nume: 'Robert Pattinson' }
    ],
    data_lansare: ISODate('2019-10-18T00:00:00.000Z'),
    boxoffice: 18.3
  },
  {
    _id: ObjectId('67af145b7c8045ea54efc0e4'),
    titlu: 'The Great Adventure',
    regizor: { id_regizor: 13, nume: 'Fey Tina' },
    actori: [
      { id_actor: 404, nume: 'Blunt Emily' },
      { id_actor: 405, nume: 'Evans Chris' }
    ],
    data_lansare: ISODate('2024-05-31T00:00:00.000Z'),
    boxoffice: 17
  }
]
FilmDB> _
```

4) In cate filme au jucat actorii? (PRIN AGREGARE)

Aceasta interogare dezintegreaza array-ul de actori astfel incat sa apara pe cate o linie, ii grupeaza dupa nume si numara de cate ori apare fiecare actor in Colectia Film (aceasta interogare putea sa fie la fel de eficient realizata si cu colectia Rol). Dupa ce numara aparitiile acestora in pune intr-o ordine descrescatoare.

```
FilmDB> db.Film.aggregate([
... { $unwind: "$actori" },
... { $group: { _id: "$actori.nume", numar_filme: { $sum: 1 } } },
... { $sort: { numar_filme: -1 } } ]).pretty();
[
  { _id: 'Willem Dafoe', numar_filme: 2 },
  { _id: 'Blunt Emily', numar_filme: 1 },
  { _id: 'Evans Chris', numar_filme: 1 },
  { _id: 'Bruce Willis', numar_filme: 1 },
  { _id: 'Robert Pattinson', numar_filme: 1 },
  { _id: 'Popescian Vio', numar_filme: 1 },
  { _id: 'Cumberbatch Benedict', numar_filme: 1 }
]
FilmDB> _
```

5) Titlul si anul lansarii filmelor(PRIN AGREGARE)

Folosind '\$project' aceasta agregare alege ce date din documentele din Film va afisa, si I-a doar anul publicarii din variabila 'data_lansare'.

```
FilmDB> db.Film.aggregate([
... { $project: { _id: 0, titlu: 1, an_lansare: { $year: "$data_lansare" } } }
... ]).pretty();
[
  { titlu: 'The Lighthouse', an_lansare: 2019 },
  { titlu: 'The Wonderful story of Henry Sugar', an_lansare: 2023 },
  { titlu: 'The Great Adventure', an_lansare: 2024 },
  { titlu: 'Haunted Night', an_lansare: 2005 }
]
FilmDB>
```

Multumesc!